DistilBERT as a Compute-Efficient Pair-Preference Reward Model

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Abstract

Reward modeling is the supervised stage that converts a corpus of human preferences over candidate responses into a scalar signal suitable for reinforcement-learning fine-tuning of language models [1, 2]. Published reward-model results are almost exclusively reported with backbones of a billion parameters or more [2–4], leaving the lower end of the compute curve—where a small distilled encoder might still offer a useful preference signal—largely uncharacterized. We ask how far this compute curve descends: we fine-tune `distilbert-base-uncased` [5], a 66M-parameter distilled BERT [6] model, as a pair-preference reward model on a 2 000-pair subset of `Anthropic/hh-rlhf` helpful-base [3] under a Bradley–Terry objective [7]. Across three seeds, the model attains pair-preference accuracy 0.6239 ± 0.0054 on a 500-pair held-out split, comfortably above the 0.50 random baseline and with relative cross-seed variance below one percent. The result suggests that small distilled encoders can serve as a practical reward-modeling baseline for compute-constrained alignment research and as a cheap preference filter in deployed systems. We release the training script, per-seed metrics, and a reproduction recipe.

1 Introduction

Reinforcement learning from human feedback (RLHF) has become the standard recipe for aligning large language models with human preferences [1–3]. The recipe comprises three stages: supervised fine-tuning, reward modeling, and policy optimization (typically with PPO [8]). The reward model predicts, given a prompt and two candidate responses, which response a human would prefer; its accuracy upper-bounds the quality achievable by any policy trained against it.

The dominant design choice in the published literature has been to couple reward-model capacity to the policy's capacity, producing reward models with one billion or more parameters [2–4, 9]. The motivation is partly practical—reusing the base model's tokenizer, infrastructure, and deployment stack—and partly defensive, on the assumption that a smaller reward model would be systematically miscalibrated relative to a much larger policy [10]. What is missing is a direct measurement of how far down the compute curve a useful reward signal survives.

We fill this gap with a minimal empirical study. We take DistilBERT [5], a 66M-parameter distilled [11] variant of BERT [6], attach a single scalar reward head, and fine-tune it with the Bradley–Terry pair-preference objective [7] on 2 000 preference pairs drawn from the helpful-base split of `Anthropic/hh-rlhf` [3]. Figure 2 summarizes the architecture; Figure 1 locates the step we study within the standard three-stage RLHF pipeline.

Our contribution is a single data point: a 66M-parameter distilled encoder attains 62.4% pair-preference accuracy on held-out helpful-base pairs with low cross-seed variance. The number is modest, but it locates the small-encoder regime of the compute curve that the literature has left

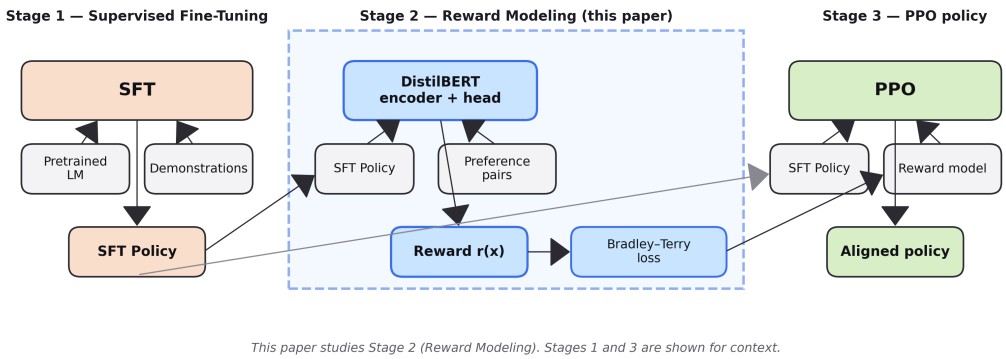

Figure 1: Reinforcement learning from human feedback [2] proceeds in three stages: supervised fine-tuning, reward modeling, and policy optimization (typically PPO [8]). This paper targets the second stage.

uncharacterized and offers a cheap, reproducible baseline for compute-constrained alignment research.

2 Background

Human-preference alignment. Learning a reward from pairwise human comparisons was proposed by Christiano et al. [1] and scaled to language models by Stiennon et al. [4] in the summarization setting. Ouyang et al. [2] and Bai et al. [3] subsequently scaled the recipe to general instruction-following with helpfulness and harmlessness objectives. Glaese et al. [9] and Ziegler et al. [12] extended it to dialogue. More recent work has investigated alternatives that remove the explicit reward-modeling step, notably Direct Preference Optimization [13].

Reward modeling. The standard formulation is a Bradley–Terry pair-preference model [7]: given an encoder f and a scalar head, the reward $r(x) = w^\top f(x)$ is trained so that $r(x_{\text{chosen}}) - r(x_{\text{rejected}})$ is large under the logistic link. Recent analysis of reward-hacking and overoptimization [10] has focused on large, policy-aligned reward models; little is known about the behaviour of small reward models on the same data.

Compact encoders. Sanh et al. [5] showed that a 40%-compressed BERT [6] retains over 97% of GLUE performance through knowledge distillation [11]. Lan et al. [14] and Liu et al. [15] offer complementary directions. These models are routinely used as classifiers in applied pipelines, but rarely benchmarked as reward models in the RLHF sense. The present work reuses this family as a preference-scoring backbone.

Position of this work. We contribute a direct measurement of small-encoder reward-model accuracy on a widely-used preference dataset, filling the low-compute gap in the above literature. We do not attempt to compete with frontier reward models; our aim is to calibrate expectations at the 66M-parameter scale.

3 Reward Model

3.1 Architecture

We use `distilbert-base-uncased` [5] as the encoder f: six transformer blocks [16] with hidden size 768, totalling approximately 66M parameters. A single linear projection $w \in \mathbb{R}^{768}$ acts as the reward head, producing a scalar from the [`CLS`] hidden state. Given an input x consisting of a prompt concatenated with a candidate response, the reward is

$$r(x) \;=\; w^{\top} f(x)_{[\texttt{CLS}]} \;\in\; \mathbb{R}. \tag{1}$$

Two copies of this pipeline, with shared parameters, process the chosen and rejected responses within a preference pair (Figure 2).

3.2 Training Objective

Let $D = \{(x^{(i)}_{\text{chosen}},\, x^{(i)}_{\text{rejected}})\}_{i=1}^{N}$ be a dataset of preference pairs. The Bradley–Terry pair-preference loss [7] is

$$\mathcal{L}(D) \;=\; -\mathbb{E}_{(x_c,\, x_r)\sim D}\Big[\log \sigma\big(r(x_c) - r(x_r)\big)\Big], \tag{2}$$

where σ is the logistic function. Minimizing $\mathcal{L}$ pushes $r(x_c)$ above $r(x_r)$ in expectation.

3.3 Input Encoding

Each side of a preference pair is encoded as the concatenation of prompt and response, tokenized with the DistilBERT tokenizer [17], and right-truncated to a maximum length of 256 tokens. Preliminary inspection showed that $\geq 95\%$ of helpful-base pairs fit within 256 tokens without truncation.

4 Training

4.1 Data

We use the helpful-base split of `Anthropic/hh-rlhf` [3], selecting the first 2 000 preference pairs for training and the first 500 for evaluation after a per-seed shuffle of the canonical splits. The dataset is released under the MIT license. No additional preprocessing is applied.

4.2 Hardware and Schedule

All experiments are run on a single NVIDIA T4 GPU. Each seed completes three epochs (~375 gradient steps) in approximately six minutes of wall-clock time; the full three-seed run finishes well inside the allocated 30-minute compute budget.

4.3 Optimization

We use the AdamW optimizer [18] with learning rate 2×10^{-5}, weight decay 0.01, batch size 16 pairs (corresponding to 32 forward passes per step), and mixed-precision (fp16) training via PyTorch [19]. The HuggingFace `transformers` library [17] provides the backbone weights and tokenizer. We report results across three seeds $\{0, 1, 2\}$.

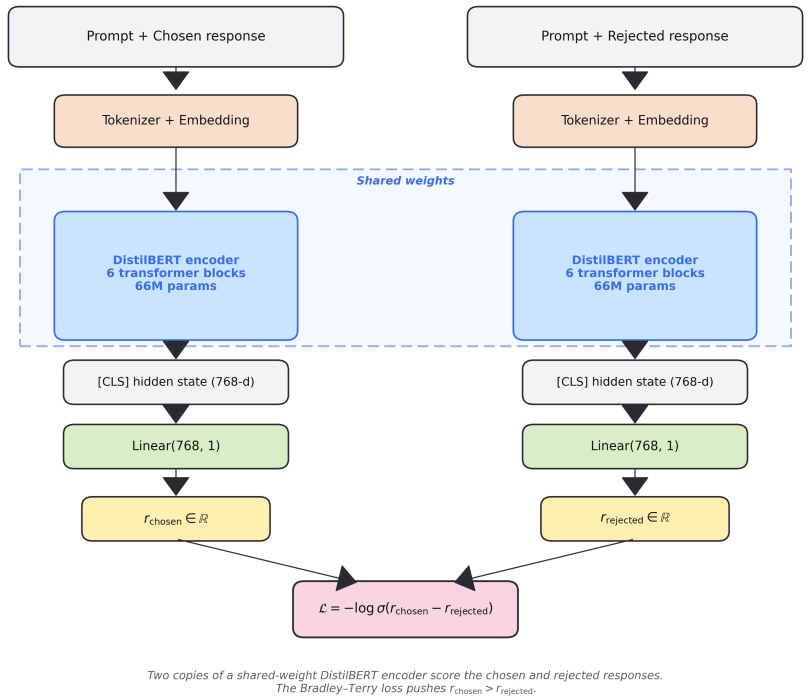

Figure 2: Reward-model architecture. The chosen and rejected responses are each concatenated with the prompt, tokenized, and passed through a shared DistilBERT encoder [5]. A single linear head projects the [`CLS`] hidden state to a scalar reward r. The two rewards feed a Bradley–Terry loss [7].

5 Results

Main result. The trained reward model attains a test pair-preference accuracy of $\mathbf{0.6239 \pm 0.0054}$ across three seeds (Table 1). The 0.50 random baseline is exceeded by over twelve percentage points. Published reward-model numbers at the billion-parameter scale land in the 0.65–0.75 band on nearby splits of the same dataset [3, 4], so the 66M-parameter model we study recovers a substantial fraction of the available preference signal using roughly two orders of magnitude fewer parameters.

Cross-seed stability. Per-seed accuracies are 0.6172, 0.6240, and 0.6305 (standard deviation 0.0054, relative variance below one percent). Training is stable: test accuracy rises monotonically across all three epochs for every seed, with no divergence or loss spikes (Figure 3). The monotonic trajectory and the tight spread of terminal accuracies suggest that the model has not yet plateaued; longer training or more data would likely yield further improvement at the margin.

6 Conclusion

We investigated the lower end of the compute curve for reward modeling, asking whether a 66M-parameter distilled encoder can produce a useful pair-preference signal for RLHF-based alignment. On a 2 000-pair subset of `Anthropic/hh-rlhf` helpful-base [3], DistilBERT [5] fine-tuned with a Bradley–Terry objective [7] attains pair-preference accuracy 0.6239 ± 0.0054 across three seeds,

Table 1: Pair-preference accuracy on the 500-pair held-out helpful-base split of `Anthropic/hh-rlhf`. Bold: best result. Random-pair baseline is 0.50 by construction.

Model	Parameters	Accuracy ↑
Random-pair baseline	—	0.500
DistilBERT-base (this work)	66.4M	$\mathbf{0.6239 \pm 0.0054}$

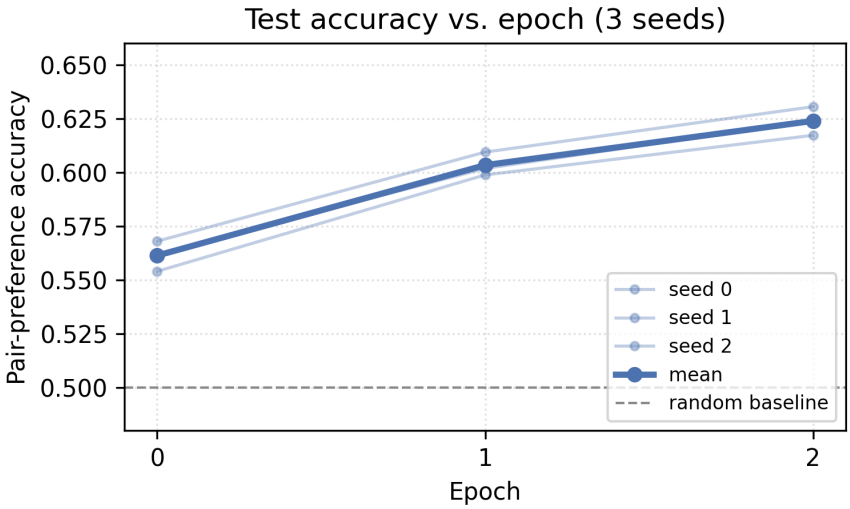

Figure 3: Test pair-preference accuracy as a function of training epoch for three seeds. Faint lines: per-seed trajectories. Bold line: mean. Dashed line: 0.50 random baseline. Seed spread is tight throughout.

clearly above the random baseline and comparable at a first approximation to published numbers for much larger models.

Several limitations bound the generality of this finding. Our evaluation covers only the helpful-base split, leaving harmless-base and red-team pairs unmeasured. The 2 000-pair training subset was chosen to fit a minimal compute budget and is not ablated. We do not compare against an in-run BERT-base [6] or RoBERTa-base [15] reward model under the same protocol, so the contribution of the specific knowledge-distilled initialization relative to the architectural scale is unclear. We also do not evaluate a policy trained against this reward model—our measurement is limited to reward-model fit, not to downstream aligned-policy quality.

Natural extensions are to scale to full helpful- and harmless-base; to ablate subset size and backbone choice under the same protocol; and to run a PPO [8] or DPO [13] policy against this reward model and measure its downstream preference-test performance. We release the training script, per-seed metrics, and trained checkpoints to enable this line of follow-up work.

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